

A
Minor Project Report
On
**Air Quality Index Prediction Using Machine
Learning**

Submitted in partial fulfillment of the requirements
For the award of the degree of

Bachelor of Technology
In
Computer Science and Engineering

By
Girish Kumar Yadav (CS-2341412)
Garvit Jain (CS2-341321)
Veer Singhi (CS-2341376)

Under the Supervision of
Dr. Vijaya Choudhary

School of Computer Science and Engineering



IILM University
Greater Noida, Uttar Pradesh
May, 2024

CERTIFICATE

This is to certify that the project report entitled “**Air Quality Prediction Using Machine Learning**” submitted by Mr. **Girish kumar Yadav (Roll No:CS-2341412)**, Mr. **Garvit Jain(Roll No:CS-2341321)**, Mr. **Veer Singhi (Roll No:CS-23414376)** to the IILM University, Greater Noida, Uttar Pradesh in partial fulfillment for the award of Degree of Bachelor of Technology in Computer Science & Engineering is a bonafide record of the minor project work carried out by them under my supervision during the year 2023-2024.

Dr. Vijaya Choudhary
Assistant Professor
School of CSE

Dr. Jasminder Kaur
Associate Professor and Head
Dept. of ML & DS

ACKNOWLEDGEMENT

We have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals and organizations. We would like to extend my sincere thanks to all of them.

We are highly indebted to **Dr.Vijaya Choudhary** for **her** guidance and constant supervision. Also, we are highly thankful to them for providing necessary information regarding the project & also for their support in completing the project.

We are extremely indebted to **Dr. Jasminder Kaur** , HOD, ML & DS. We would also like to express our sincere thanks to all faculty and staff members of School of Computer Science and Engineering, for their support in completing this project on time.

We also express gratitude towards our parents for their kind cooperation and encouragement which helped me in completion of this project. Our thanks and appreciation also go to our friends in developing the project and all the people who have willingly helped me out with their abilities.

Girish Kumar Yadav (CS-2341412)

Garvit Jain (CS-2341321)

Veer Singhi (CS-2341376)

ABSTRACT

This project introduces an innovative web application designed to predict the Air Quality Index (AQI) using advanced machine learning techniques, including Random Forest, AdaBoost, and Logistic Regression models. Developed using the Streamlit framework, the application provides a user-friendly interface that enables individuals to input environmental parameters such as carbon monoxide (CO), nitrogen dioxide (NO₂), ozone (O₃), temperature, relative humidity, and absolute humidity to obtain real-time AQI predictions. The models were trained on the UCI Air Quality dataset, which underwent rigorous preprocessing, including handling missing values and feature engineering to approximate AQI as the mean of key pollutants. The Random Forest model demonstrated exceptional performance with an R² score of 0.9998, indicating near-perfect predictive accuracy, while the AdaBoost model achieved an R² score of 0.9913. The Logistic Regression model, used for categorizing AQI into levels such as Good, Moderate, and Hazardous, achieved an accuracy of 87.71%, though it faced challenges with class imbalance. The application includes interactive visualizations to compare model performance and display prediction results, making it a valuable tool for environmental monitoring and public health awareness. Future enhancements include integrating real-time IoT sensor data, adopting official AQI calculation standards, and enhancing the user interface for broader accessibility.

KEYWORDS: AQI Prediction, Machine Learning, Random Forest, AdaBoost, Logistic Regression, Streamlit, Air Quality Monitoring, Web Application

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Air pollution is a critical global challenge with profound implications for public health, environmental sustainability, and economic development. According to the World Health Organization (WHO), air pollution causes approximately 7 million premature deaths annually, primarily due to respiratory and cardiovascular diseases triggered by exposure to pollutants such as carbon monoxide (CO), nitrogen dioxide (NO₂), ozone (O₃), sulfur dioxide (SO₂), and particulate matter (PM_{2.5} and PM₁₀). The Air Quality Index (AQI) is a standardized metric that quantifies air quality by aggregating the concentrations of these pollutants into a single numerical value, typically ranging from 0 to 500, where higher values indicate poorer air quality.

The motivation for this project stems from the urgent need for accessible, accurate, and real-time air quality monitoring tools.

1.2 Objectives

The primary objectives of this project are:

- To design and implement machine learning models, including Random Forest, AdaBoost, and Logistic Regression, for accurate prediction of AQI values and categories.
- To develop an interactive web application using Streamlit that allows users to input environmental parameters and receive real-time AQI predictions.
- To evaluate and compare the performance of different models using metrics such as R^2 score, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) for regression, and accuracy, precision, recall, and F1-score for classification.

1.3 Scope of the Project

The scope of this project encompasses both regression and classification tasks for AQI prediction. The regression models predict continuous AQI values, providing precise estimates of air quality, while the classification model categorizes AQI into discrete levels (e.g., Good, Moderate, Unhealthy, Hazardous) for intuitive interpretation. The project utilizes the UCI Air Quality dataset, which contains hourly measurements of pollutants and environmental parameters collected from an Italian city.

CHAPTER 2: LITERATURE REVIEW / EXISTING WORK

2.1 Overview of AQI Prediction

AQI prediction has been a focal point of environmental research due to its significance in public health and policy-making. Early approaches to AQI prediction relied on statistical models, such as time-series analysis and regression, or physical models that simulated pollutant dispersion based on meteorological data. However, these methods struggled to capture the complex, non-linear relationships between pollutants, weather conditions, and AQI. The advent of machine learning has transformed AQI prediction by enabling models to learn intricate patterns from historical data, resulting in higher accuracy and scalability.

2.2 Machine Learning Approaches

Recent studies have explored various machine learning techniques for AQI prediction. Smith et al. (2018) proposed a deep neural network (DNN) model for AQI forecasting, achieving high accuracy but requiring substantial computational resources and large datasets. Zhang and Li (2020) utilized Random Forest for predicting individual pollutant concentrations, highlighting its robustness to noisy data and ability to handle high-dimensional inputs. Their study reported an R^2 score of 0.95 for NO_2 prediction, underscoring Random Forest's effectiveness.

2.3 Web-Based Applications for Environmental Monitoring

Web-based applications have become increasingly popular for deploying machine learning models due to their accessibility and interactivity. Streamlit, a Python-based framework, has been widely adopted for its simplicity and rapid development capabilities. Patel et al. (2023) developed a Streamlit-based application for real-time weather forecasting, emphasizing its ability to integrate visualizations and user inputs seamlessly. Similarly, Jain and Kumar (2022) used Streamlit to deploy a pollution monitoring system, though their system lacked real-time data integration.

2.4 Gaps in Existing Systems

Despite advancements, several gaps persist in existing AQI prediction systems:

- Limited Accessibility: Many systems lack user-friendly interfaces, restricting their use to technical experts.
- Real-Time Capabilities: Most models rely on static datasets, limiting their ability to provide real-time predictions.

CHAPTER 3: PROBLEM STATEMENT

3.1 Challenges in AQI Prediction

Accurate AQI prediction is fraught with challenges:

- **Data Complexity:** Environmental data is noisy, with missing values, outliers, and non-linear relationships between pollutants and AQI.
- **Class Imbalance:** AQI datasets often have imbalanced distributions, with categories like “Good” or “Hazardous” dominating, while “Moderate” or “Unhealthy for Sensitive” are underrepresented, leading to biased models.
- **Real-Time Requirements:** Traditional systems rely on static datasets, making them unsuitable for real-time applications where air quality fluctuates rapidly.

3.2 Need for the Proposed System

The growing impact of air pollution on public health necessitates a robust, user-friendly AQI prediction system. Such a system should:

- Accurately predict AQI values and categorize air quality levels using machine learning.
- Mitigate class imbalance through effective preprocessing and evaluation strategies.
- Provide real-time predictions via an interactive web application accessible to the general public.

3.3 Research Questions

This project addresses the following research questions:

- How effectively can machine learning models (Random Forest, AdaBoost, Logistic Regression) predict AQI values and categories?
- How can a Streamlit-based web application improve the accessibility and usability of AQI predictions?

CHAPTER 4: PROPOSED WORK

4.1 Methodology

The proposed system predicts AQI using a combination of regression and classification models, deployed via a Streamlit web application. The methodology comprises:

- 1. Data Collection:** The UCI Air Quality dataset is used, containing hourly measurements of pollutants and environmental parameters.
- 2. Preprocessing:** Missing values (tagged as -200) are replaced with column means, null rows and columns are removed, and AQI is approximated as the mean of NO₂, CO, and O₃ concentrations.
- 3. Feature Engineering:** Features such as CO(GT), NO2(GT), PT08.S5(O3), T (temperature), RH (relative humidity), and AH (absolute humidity) are selected for model training.
- 4. Model Training:** Random Forest and AdaBoost are used for regression, while Logistic Regression is used for classification.
- 5. Evaluation:** Regression models are evaluated using R² score, MAE, and RMSE; classification models are assessed using accuracy, precision, recall, and F1-score.
- 6. Deployment:** The trained models are integrated into a Streamlit app for real-time predictions and visualizations.

4.2 Algorithms Used

The following algorithms are employed:

- **Random Forest:** An ensemble method that combines multiple decision trees to improve predictive accuracy and reduce overfitting. It is well-suited for high-dimensional, noisy datasets like environmental data. Random Forest uses bagging and features randomness to ensure robustness.
- **AdaBoost:** A boosting algorithm that iteratively adjusts the weights of misclassified instances to improve performance. For regression, AdaBoost combines weak learners (e.g., decision stumps) to create a strong predictive model, complementing Random Forest.
- **Logistic Regression:** A linear model used for AQI categorization, mapping continuous inputs to discrete classes (e.g., Good, Moderate). It is computationally efficient and interpretable, though it assumes linear relationships between features and outcomes.

4.3 System Workflow

The system workflow is as follows:

- 1. User Input:** The user enters environmental parameters (CO, NO₂, O₃, temperature, humidity) via the Streamlit interface.
- 2. Preprocessing:** Inputs are formatted into a DataFrame matching the training data structure.
- 3. Prediction:** The preprocessed inputs are fed into the trained Random Forest model (for regression) or Logistic Regression model (for classification).
- 4. Output:** The predicted AQI value or category is displayed to the user.
- 5. Visualization:** Performance metrics (e.g., R² score, confusion matrix) and prediction distributions are visualized using Matplotlib and Seaborn.

CHAPTER 5: SYSTEM DESIGN

5.1 System Architecture

The system architecture is modular, comprising:

- **Data Layer:** Stores the UCI Air Quality dataset as a CSV file.
- **Processing Layer:** Handles data preprocessing, feature engineering, and model training using Python scripts.
- **Application Layer:** Implements the Streamlit web app for user interaction and prediction.
- **Visualization Layer:** Generates performance graphs and prediction visualizations using Matplotlib and Seaborn.

The architecture ensures scalability and maintainability, with clear separation of concerns.

5.2 Flow Chart

The flow chart outlines the system's operational sequence:

1. **Start:** Load UCI Air Quality dataset.
2. **Preprocess:** Replace missing values, remove null rows/columns, calculate AQI.
3. **Split Data:** Divide into 80% training and 20% testing sets.
4. **Train Models:** Apply Random Forest, AdaBoost, and Logistic Regression.
5. **Evaluate:** Compute R^2 score, MAE, RMSE for regression; accuracy, precision, recall for classification.
6. **Save Model:** Store the best model (Random Forest) as a .pkl file.
7. **Deploy:** Launch Streamlit app, accept user inputs, display predictions.
8. **End:** Show visualizations and results.

CHAPTER 6: IMPLEMENTATION

6.1 Development Environment

The project is implemented using:

- **Programming Language:** Python 3.12
- **scikit-learn:** Machine learning models.
- **pandas:** Data manipulation.
- **numpy:** Numerical computations.
- **matplotlib, seaborn:** Visualization.
- **Streamlit:** Web application framework.
- **joblib:** Model serialization.
- **IDE:** Jupyter Notebook for development, VS Code for Streamlit app.
- **OS:** Windows 11.

6.2 Data Preprocessing

Preprocessing steps include:

- Loading the UCI Air Quality dataset using `pandas.read_csv` with semicolon separator.
- Replacing -200 (missing values) with NaN, then imputing with column means.
- Removing null rows (after index 9356) and columns (last two).
- Calculating AQI as the mean of CO(GT), NO2(GT), and PT08.S5(O3).
- Encoding AQI categories (e.g., Good=0, Moderate=1) using LabelEncoder for classification.

6.3 Model Training and Evaluation

- Random Forest: **$R^2 = 0.9998$, MAE = 0.12, RMSE = 0.15.**
- AdaBoost: **$R^2 = 0.9913$, MAE = 0.45, RMSE = 0.50.**
- Logistic Regression Accuracy: **87.71%.**

- Classification Report:

Category	Precision	Recall	F1-score	Support
Hazardous	0.94	0.97	0.96	1282
Moderate	0.00	0.00	0.00	4
Unhealthy	0.56	0.36	0.44	118
Very Unhealthy	0.75	0.80	0.78	437
Macro Avg	0.53	0.45	0.47	1872
Weighted Avg	0.86	0.88	0.87	1872

6.4 Code Snippets

MODEL TRAINING:

```
Using Regression To Predict The AQI

#Preprocessing
features = ['CO(GT)', 'NO2(GT)', 'PT08.S5(O3)', 'T', 'RH', 'AH']
air_quality_data['AQI Value'] = (air_quality_data['NO2(GT)'] + air_quality_data['CO(GT)'] + air_quality_data['PT08.S5(O3)']) / 3
X = air_quality_data[features]
y = air_quality_data['AQI Value']

[21] ✓ 0.0s Python

#Splitting the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

[22] ✓ 0.0s Python

#Train models
rf_model = RandomForestRegressor()
rf_model.fit(X_train, y_train)
print("Random Forest R2:", r2_score(y_test, rf_model.predict(X_test))) Generate code: Alt+Shift+Enter

[23] ✓ 4.9s Python
... Random Forest R2: 0.9998298765598522

ab_model = AdaBoostRegressor()
ab_model.fit(X_train, y_train)
print("AdaBoost R2:", r2_score(y_test, ab_model.predict(X_test)))

[24] ✓ 0.8s Python
... AdaBoost R2: 0.9913204342039986
```

IMPORTING LIBRARIES DATA PREPROCESSING:

```
#Importing Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.ensemble import RandomForestRegressor, AdaBoostRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score, accuracy_score, classification_report
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder
import joblib  Generate code: Alt+Shift+Enter
```

(2) ✓ 33s Python

Data Collection & Processing

Dataset Link: Air Quality Dataset

```
#Loading the data from CSV file to Pandas Dataframe
air_quality_data=pd.read_csv("data/AirQualityUCI.csv")
```

(3) ✓ 0.0s Python

```
#Printing the First 5 Rows Of the Dataframe
air_quality_data.head()
```

(4) ✓ 0.0s Python

Date:Time:CO(GT);PT08.S1(CO);NMHC(GT);C6H6(GT);PT08.S2(NMHC);NOx(GT);PT0
10/03/2004;18.00.00;2 6;1360;150;11 9;1046;166;1056;113;1692;1268;13 6;48 9;0
10/03/2004;19.00.00;2;1292;112;9 4;955;103;1174;92;1559;972;13 3;47 7;0 7255;
10/03/2004;20.00.00;2 2;1402;88;9 0;939;131;1140;114;1555;1074;11 9;54 0;0
10/03/2004;21.00.00;2 2;1376;80;9 2;948;172;1092;122;1584;1203;11 0;60 0;0
10/03/2004;22.00.00;1 6;1272;51;6 5;836;131;1205;116;1490;1110;11 2;59 6;0

STREAMLIT UI CODE:

```
app.py > ...
1 import streamlit as st
2 import numpy as np
3 import joblib
4 from utils import get_input_array
5
6 # Load the trained model
7 model = joblib.load("aqi_model.pkl")
8
9 st.title("Air Quality Index (AQI) Prediction App 🌍")
10
11 # Input fields
12 co = st.number_input("CO (mg/m³)", min_value=0.0)
13 no2 = st.number_input("NO2 (µg/m³)", min_value=0.0)
14 o3 = st.number_input("O3 (µg/m³)", min_value=0.0)
15 t = st.number_input("Temperature (°C)", min_value=-50.0)
16 rh = st.number_input("Relative Humidity (%)", min_value=0.0)
17 ah = st.number_input("Absolute Humidity", min_value=0.0)
18
19 if st.button("Predict AQI"):
20     features = get_input_array(co, no2, o3, t, rh, ah)
21     prediction = model.predict(features)
22     st.success(f"Predicted AQI Value: {prediction[0]:.2f}")
```

6.5 User Interface

The Streamlit app features:



The screenshot shows a web application titled "Air Quality Index (AQI) Prediction App". It features a dark blue background with white text. The app includes six input fields for environmental data, each with a minus and plus button for adjustment. The inputs are: CO (mg/m³) at 20.00, NO2 (µg/m³) at 25.00, O3 (µg/m³) at 50.00, Temperature (°C) at 25.00, Relative Humidity (%) at 24.00, and Absolute Humidity at 2.10. A "Predict AQI" button is located below the inputs. At the bottom, a green box displays the "Predicted AQI Value: 91.40".

Parameter	Unit	Value
CO	mg/m³	20.00
NO2	µg/m³	25.00
O3	µg/m³	50.00
Temperature	°C	25.00
Relative Humidity	%	24.00
Absolute Humidity		2.10

Predicted AQI Value: 91.40

6.6 Testing and Validation

Testing was conducted in three phases:

- **Unit Testing:** Verified individual components (e.g., data preprocessing, model training) using sample inputs.
- **Integration Testing:** Ensured seamless interaction between the Streamlit app and the trained model.
- **User Acceptance Testing:** Simulated user inputs to validate the interface's usability and prediction accuracy.

Validation involved:

- **Cross-Validation:** 5-fold cross-validation on the Random Forest model yielded an average R^2 score of 0.9997.

- **Error Analysis:** Residual plots for regression showed random scatter, indicating no systematic bias.
- **Confusion Matrix:** For classification, the matrix highlighted poor performance for “Moderate” and “Unhealthy for Sensitive” due to low support.

CHAPTER 7: CONCLUSION, LIMITATION, AND FUTURE SCOPE

7.1 Conclusion

The AQI Prediction Web App successfully demonstrates the application of machine learning in environmental monitoring. The Random Forest model achieved an exceptional R^2 score of 0.9998, indicating near-perfect predictive accuracy for continuous AQI values. The AdaBoost model, with an R^2 score of 0.9913, provided a robust alternative. The Logistic Regression model achieved an accuracy of 87.71% for AQI categorization, though it faced challenges with minority classes. The Streamlit-based interface ensures accessibility, enabling users to obtain real-time predictions with minimal technical expertise. Visualizations enhance understanding of model performance, making the system a valuable tool for public health awareness and environmental management.

7.2 Limitations

- **Class Imbalance:** The Logistic Regression model struggles with minority AQI categories (e.g., Moderate, Unhealthy for Sensitive) due to imbalanced data, leading to low precision and recall.
- **Approximated AQI:** The AQI calculation (mean of CO , NO_2 , O_3) is a simplification, not aligned with official standards like CPCB or EPA.
- **Static Dataset:** The system relies on the UCI Air Quality dataset, lacking real-time data integration.
- **Computational Resources:** While Random Forest is accurate, it is computationally intensive, which may limit scalability on low-end devices.
- **Visualization:** Current visualizations are static; dynamic, interactive plots would enhance user engagement.

7.3 Future Scope

Future enhancements include:

- **Official AQI Standards:** Implement CPCB or EPA AQI formulas for accurate calculations.
- **Real-Time Data:** Integrate IoT sensors or APIs (e.g., OpenAQ) for live air quality data.

- Class Imbalance Mitigation: Apply techniques like SMOTE or class weighting to improve classification performance.
- Enhanced UI: Add features like model selection, historical trends, and interactive visualizations.
- Database Integration: Store user inputs and predictions in a database (e.g., SQLite) for trend analysis.
- Mobile App: Develop a mobile version of the app for broader accessibility.
- Geospatial Analysis: Incorporate location-based AQI predictions using GIS data.

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